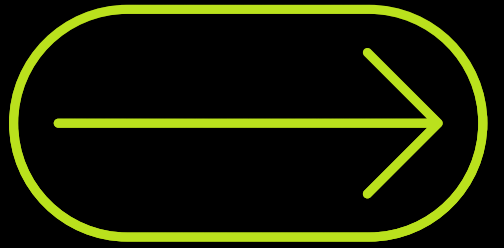


Reddit Project: Web API and NLP (Harry Potter and Marvel)

Mohona Yesmin



01 - Introduction



02 - EDA Data Visualization



03 - Model Analysis



04 - Conclusions





01 - Introduction

Problem Statement

Identify the preferences of Harry Potter and Marvel fans on social media platforms to inform targeted advertising strategies for Halloween merchandise

Marketing companies seek to better understand the preferences of Harry Potter and Marvel fans on social media to optimize targeted Halloween merchandise ads. This understanding enables more effective advertising, leading to increased engagement and conversions for Halloween sales.

01 - Data Introduction

Data set

1. *marvel.csv*
2. *harrypotter.csv*

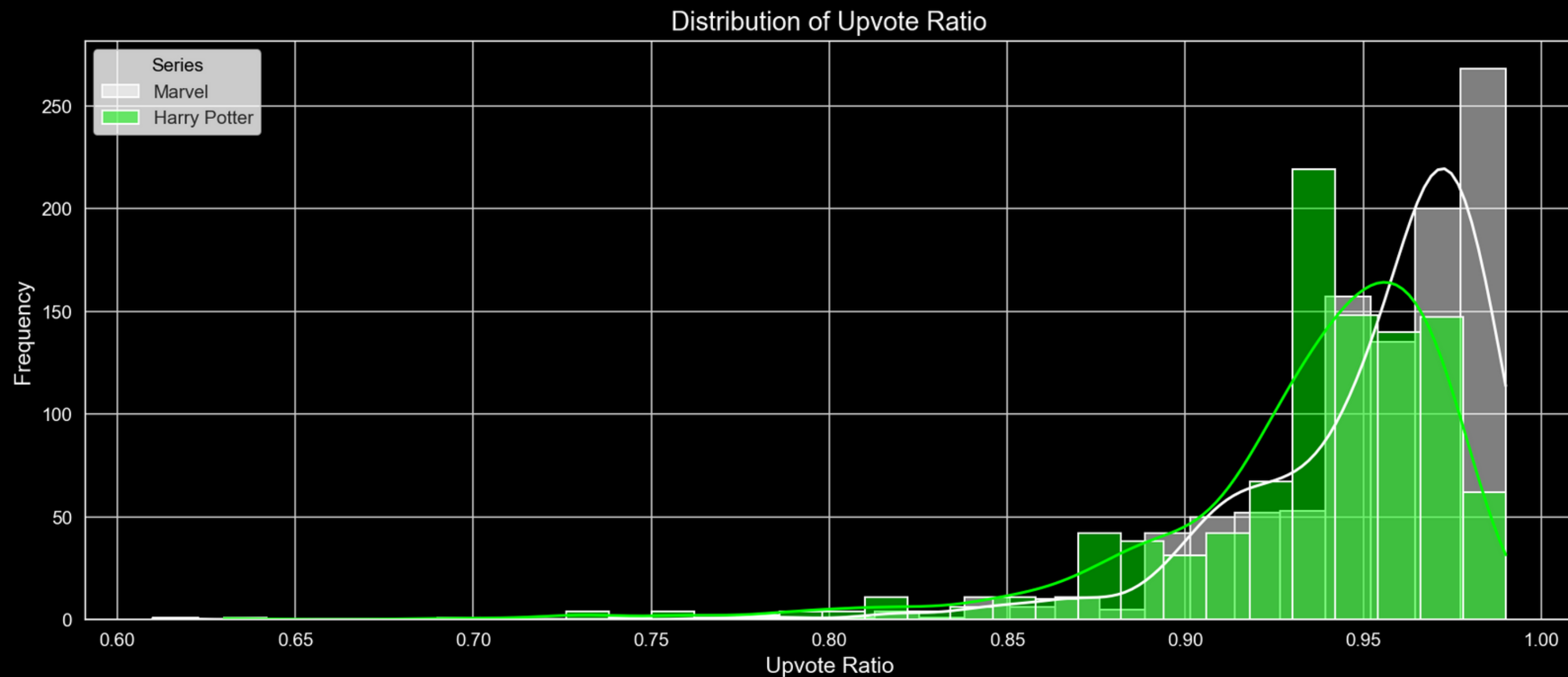
Data



1. *'id',*
2. *'title',*
3. *'content',*
4. *'score',*
5. *'num_comments',*
6. *'author',*
7. *'created_utc',*
8. *'upvote_ratio',*
9. *'subreddit_flair',*
10. *'submission_datetime',*
11. *'post_type',*
12. *'entity_recognition',*
13. *'text_sentiment'*

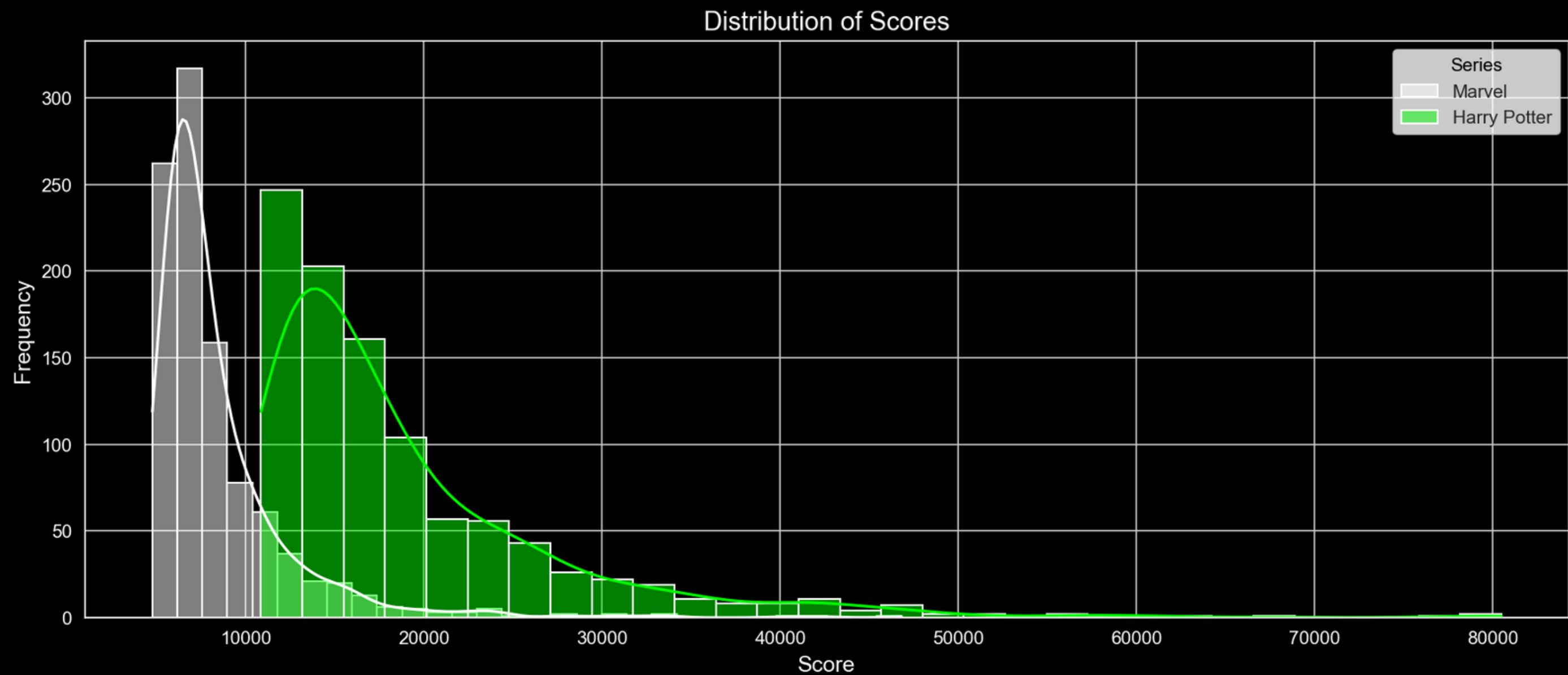
02 - Data Visualization

Distribution of Upvote Ratio - Marvel vs. Harry Potter



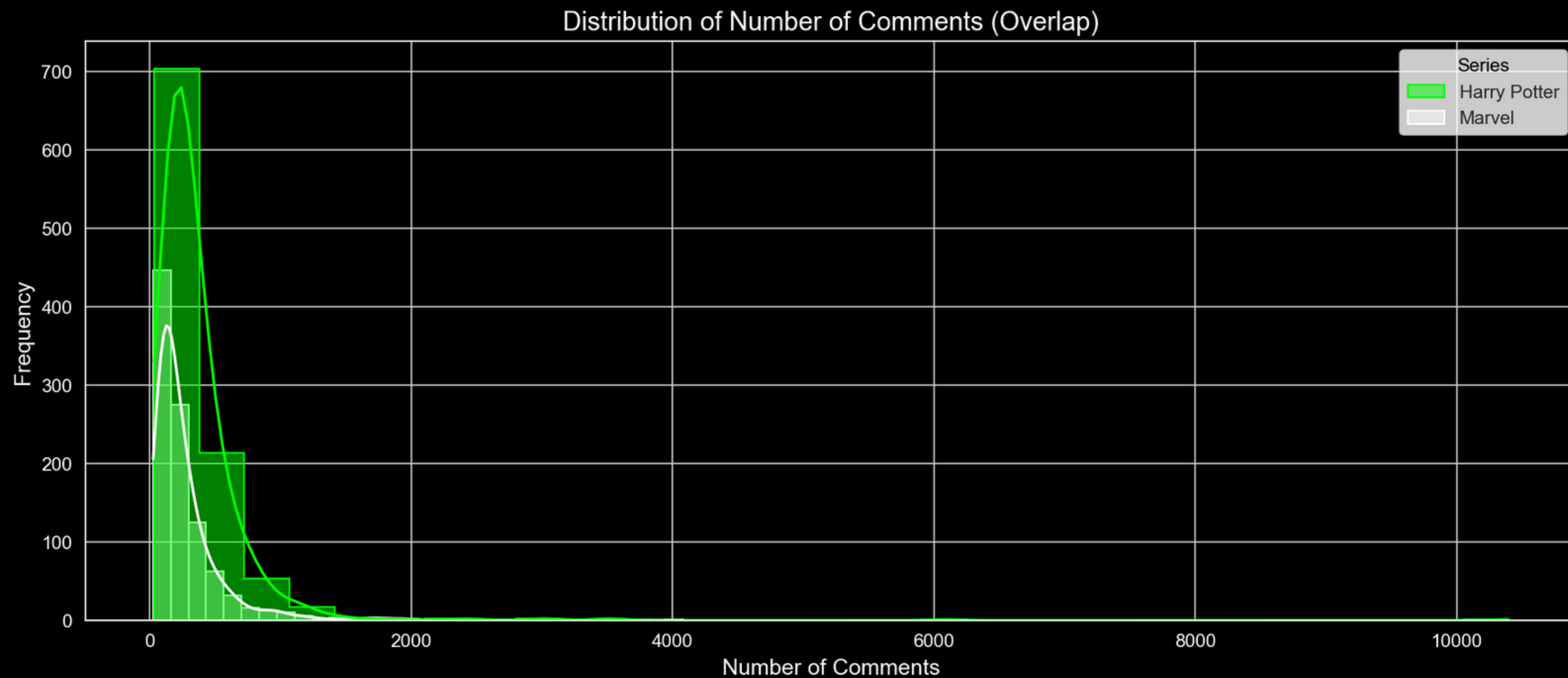
02 - Data Visualization

Distribution of Scores - Marvel vs. Harry Potter



02 - Data Visualization

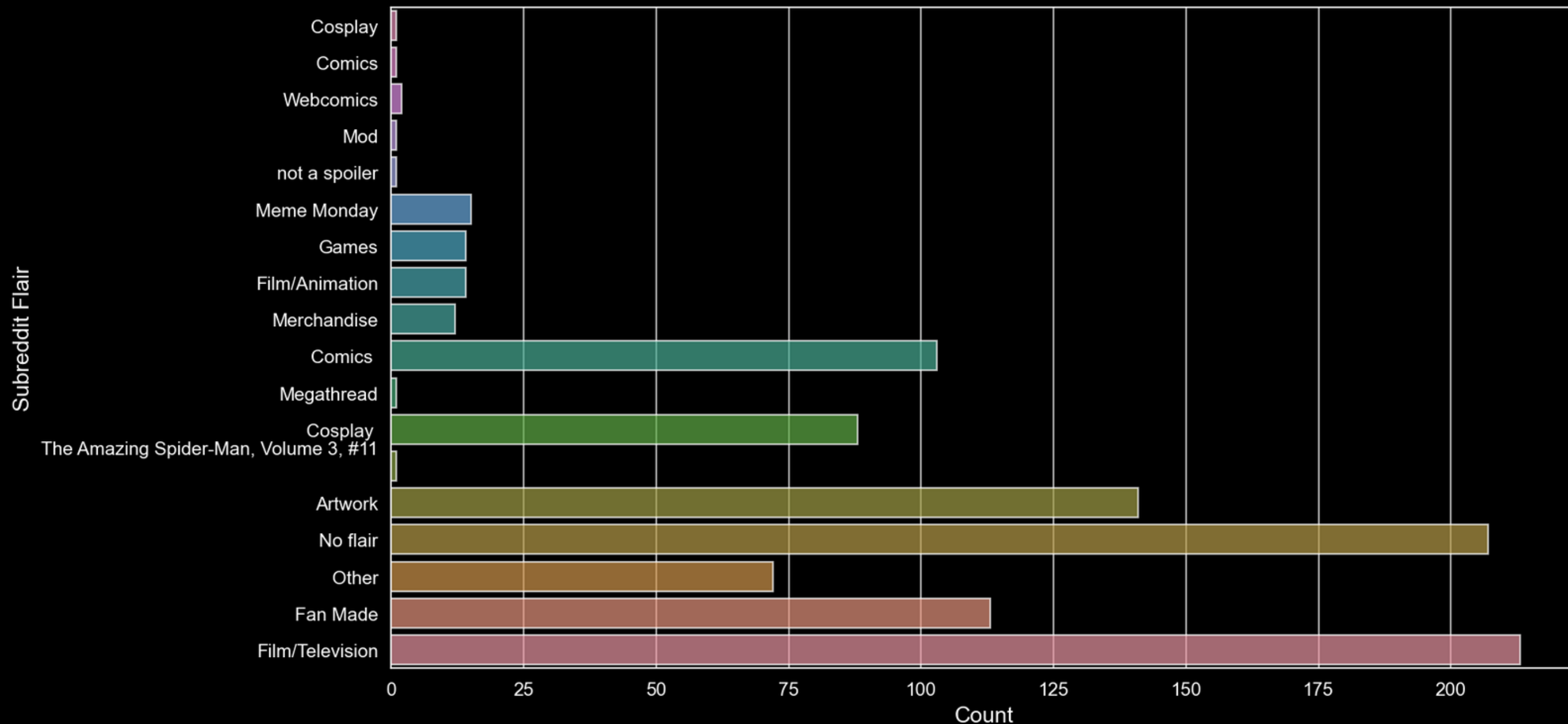
Distribution of Number of Comments - Marvel vs. Harry Potter



02 - Data Visualization

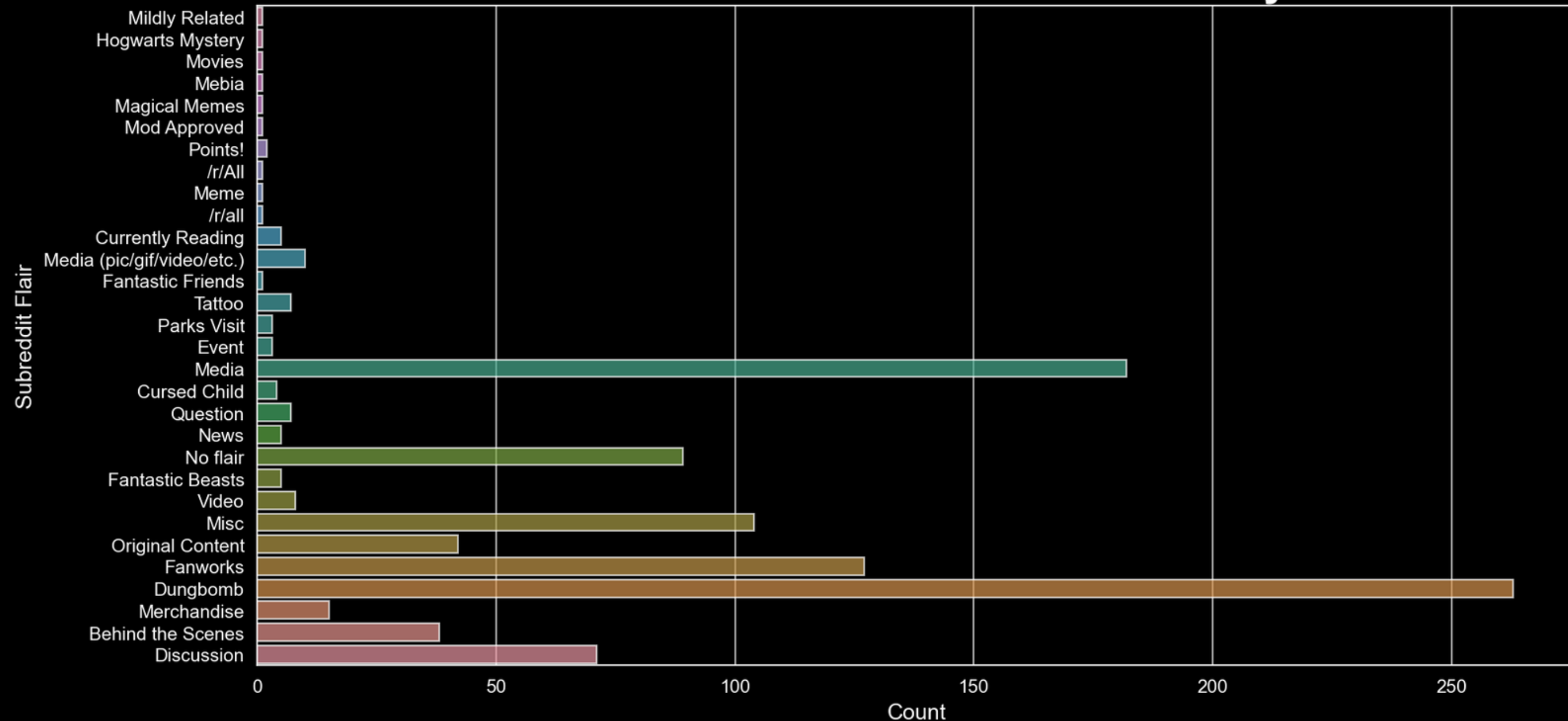


Count of Different Subreddit Flairs - Marvel



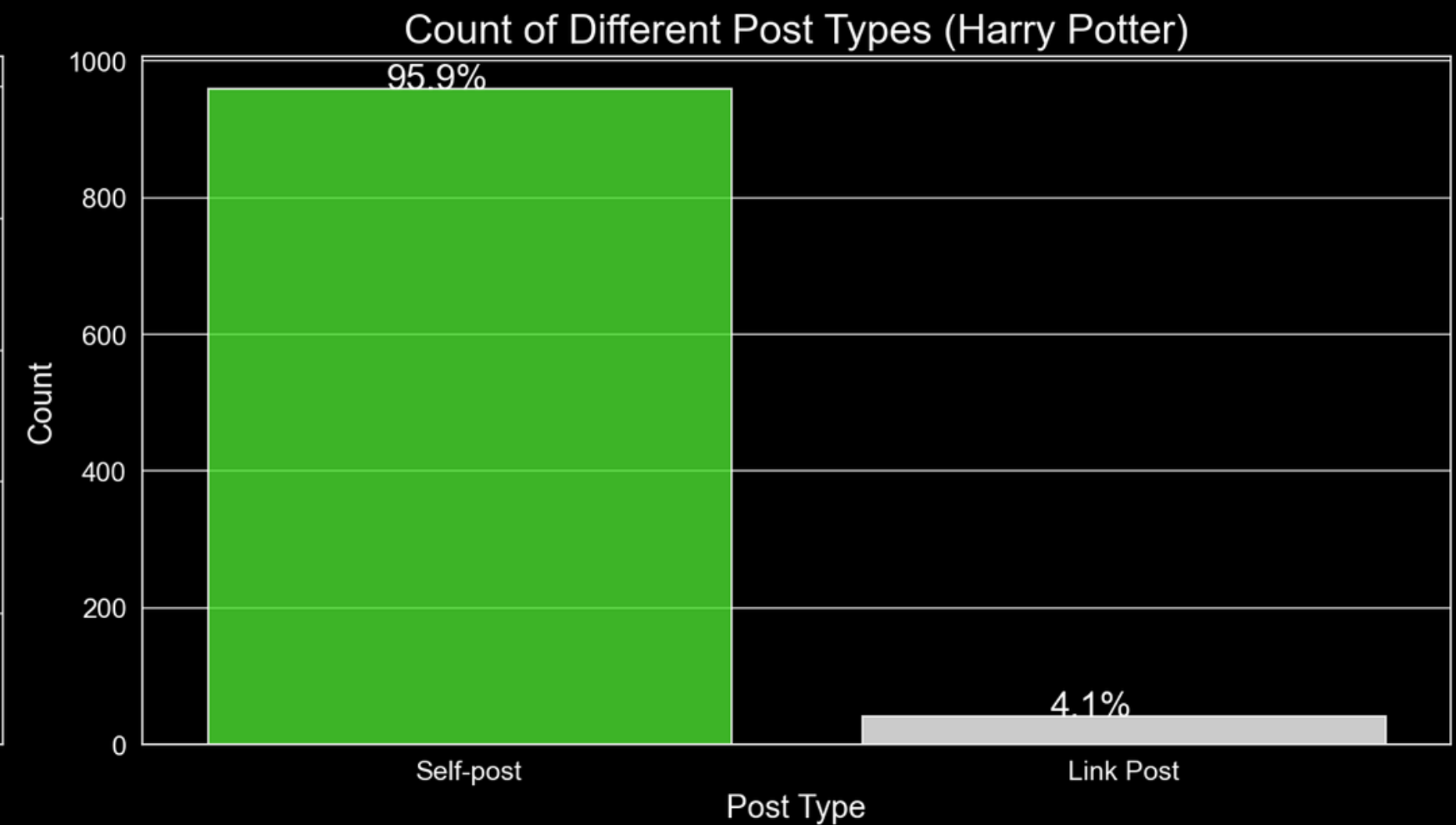
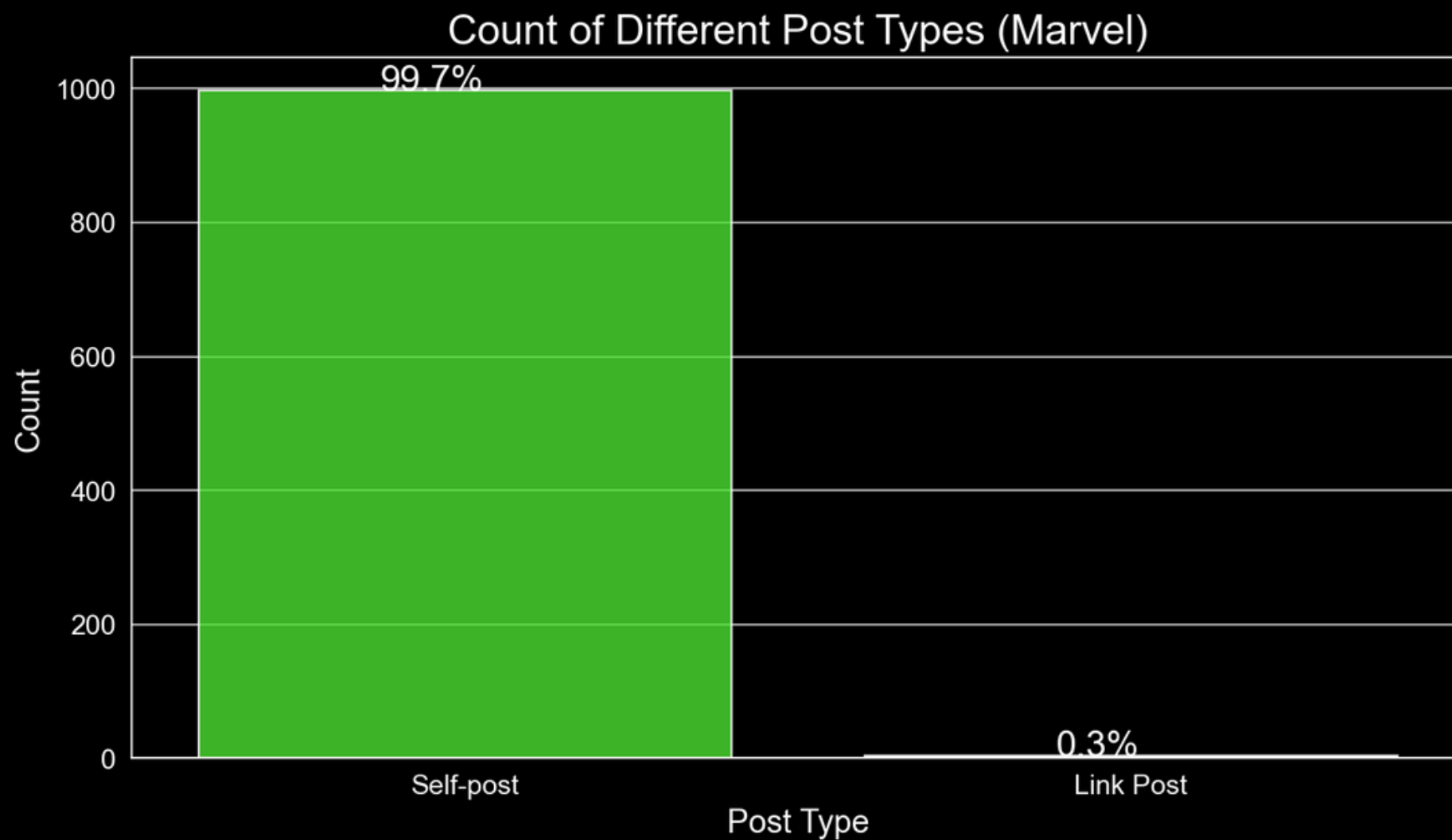
02 - Data Visualization

Count of Different Subreddit Flairs - Harry Potter



02 - Data Visualization

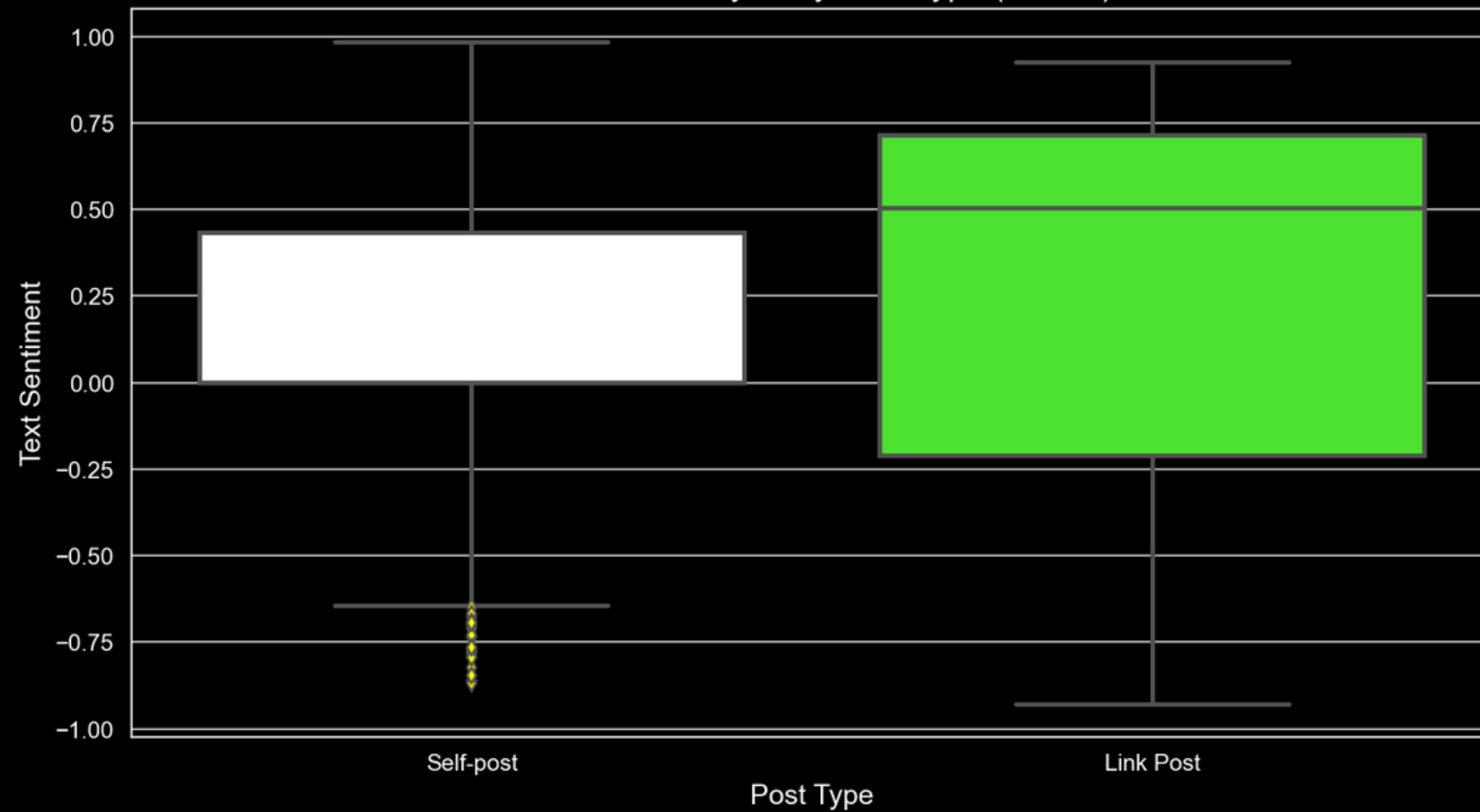
Count of Different Post Types - Marvel vs. Harry Potter



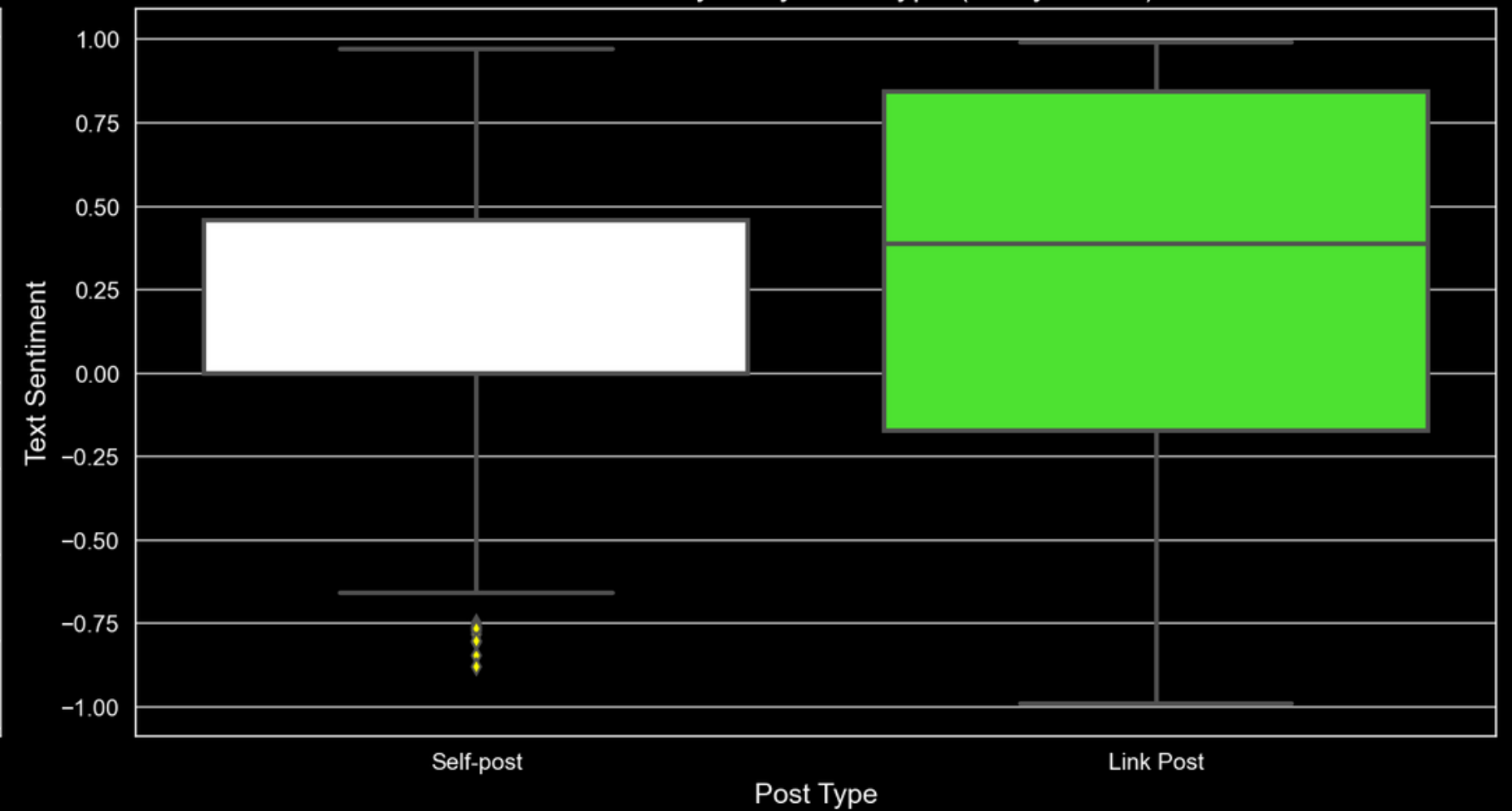
02 - Data Visualization

Sentiment Analysis by Post Type - Marvel vs. Harry Potter

Sentiment Analysis by Post Type (Marvel)



Sentiment Analysis by Post Type (Harry Potter)



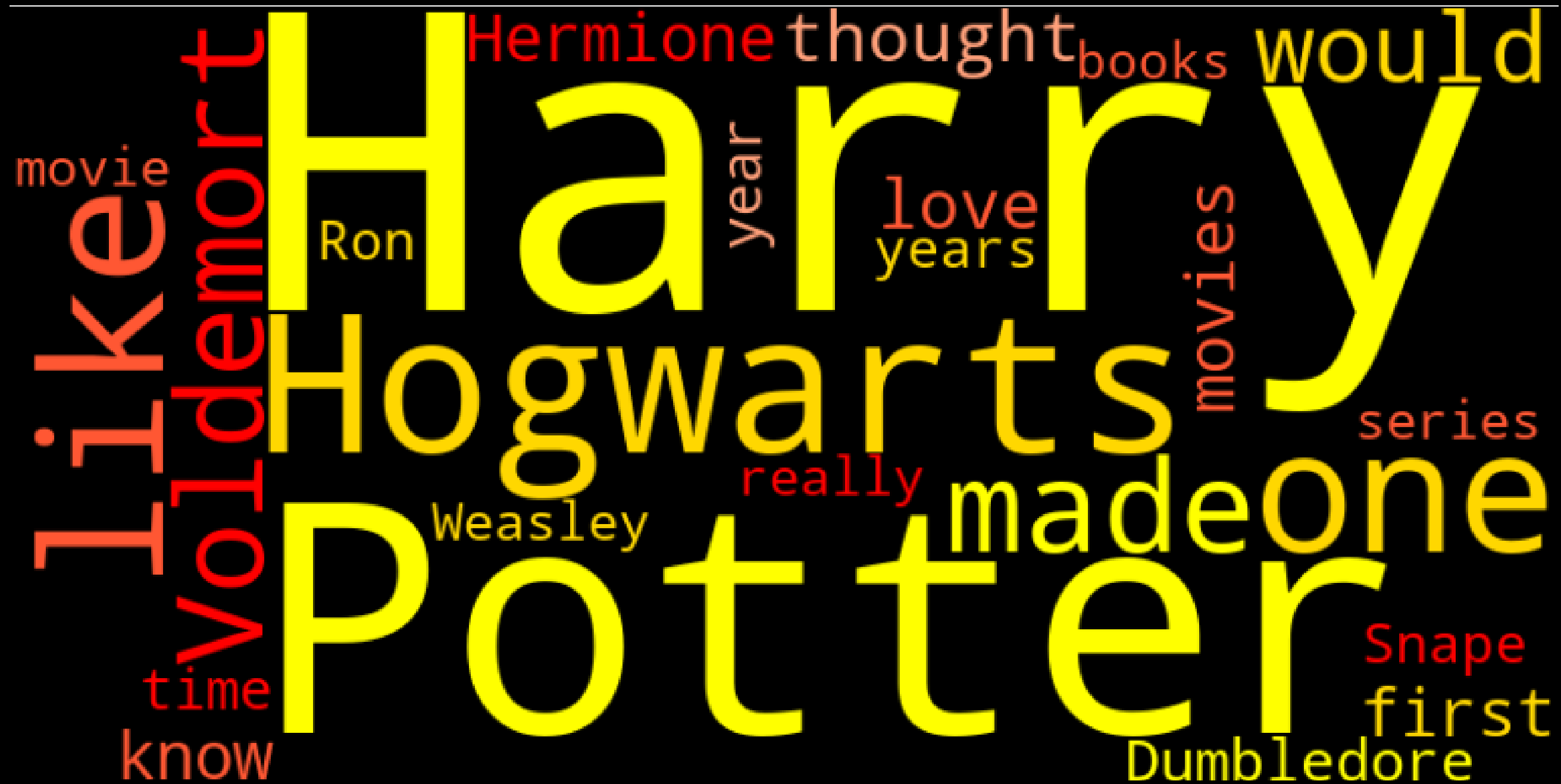
Common Words

Data Columns

- 1. Tittle*
- 2. Content*

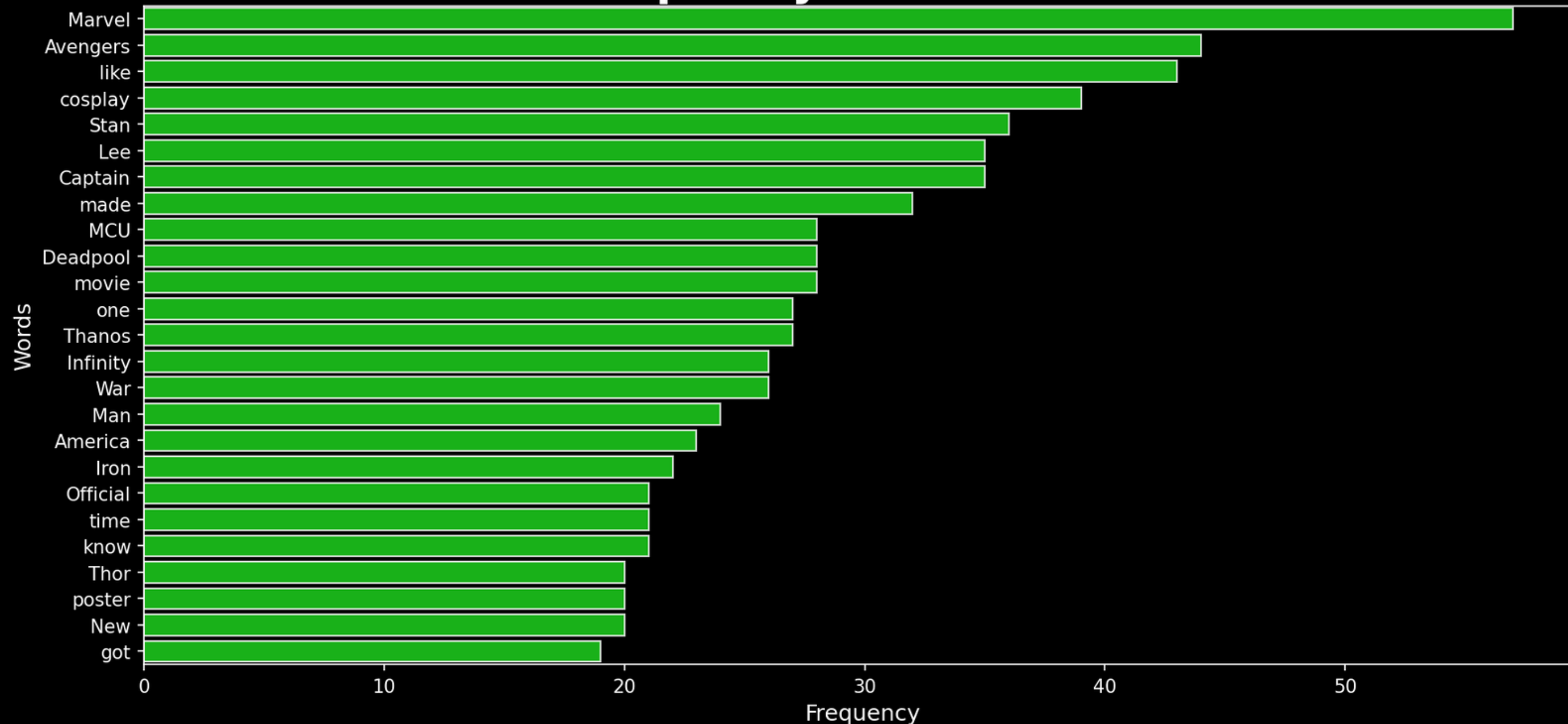
Harry Potter

Common Words in Title and Content



02 - Data Visualization

Word Frequency Horizontal Bar Plot



1. Marvel
2. Avengers
3. like
4. cosplay
5. Stan

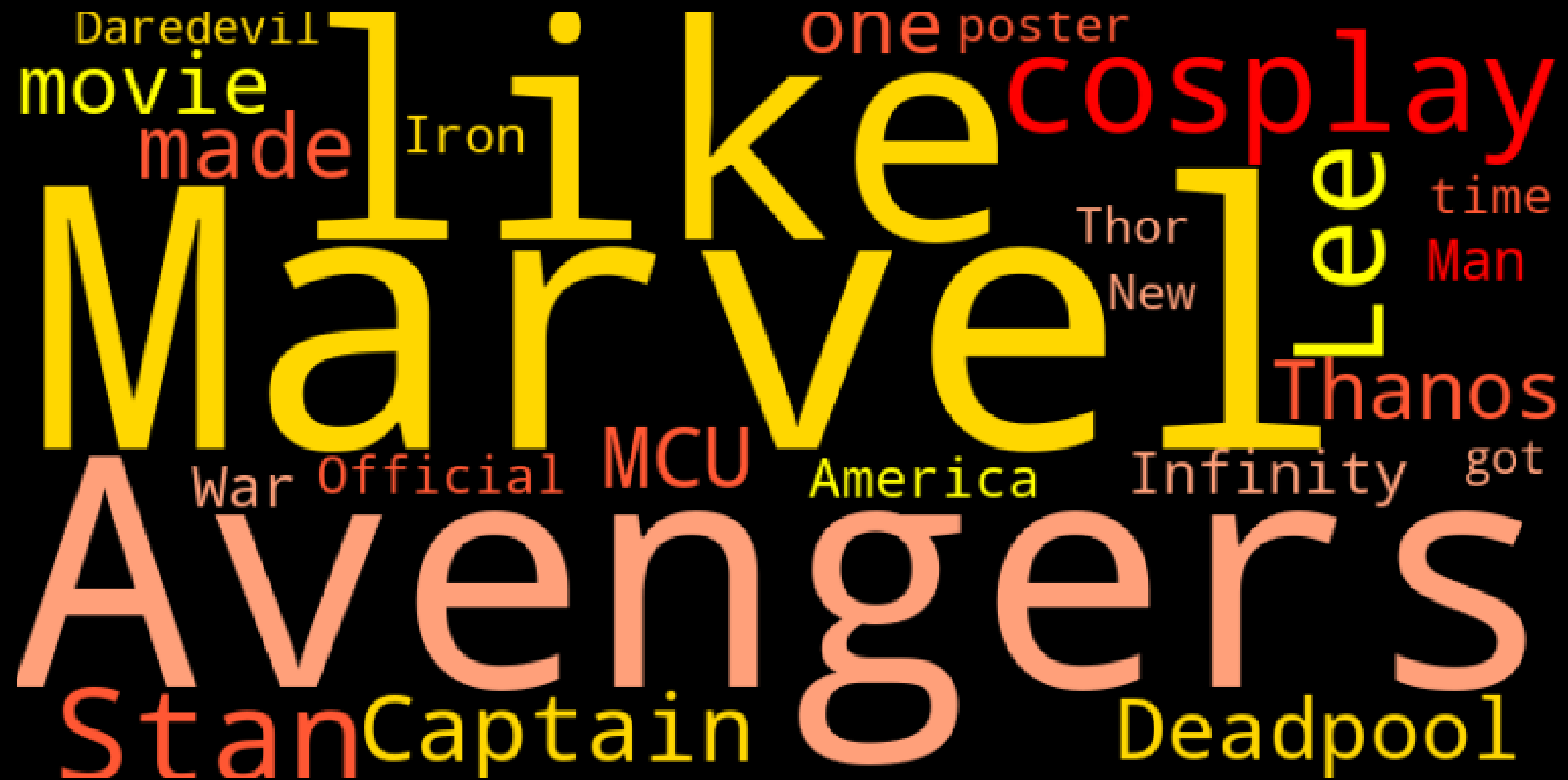
Common Words

Data Columns

- 1. Tittle*
- 2. Content*

Marvel

Common Words in Tittle

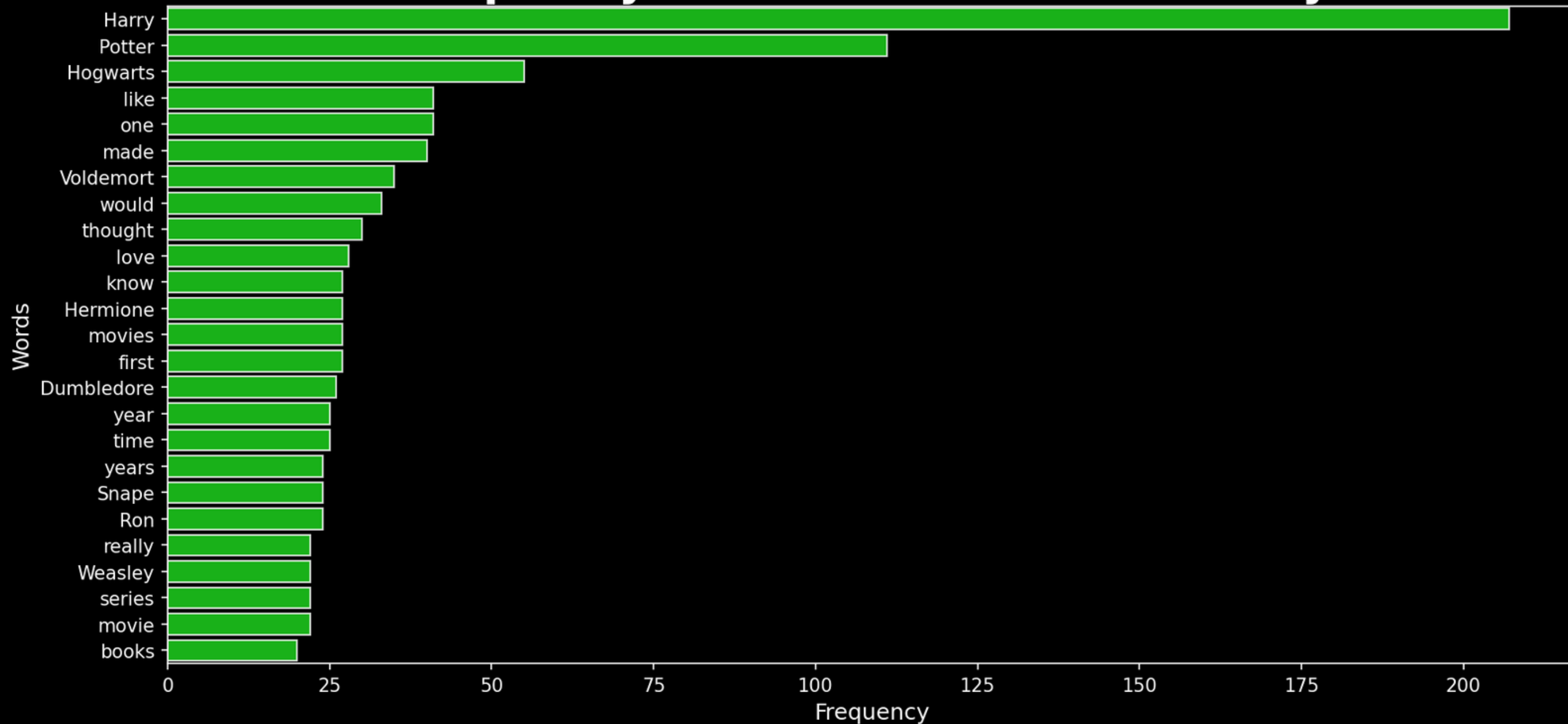


A word cloud featuring various terms related to the Marvel Cinematic Universe. The words are arranged in a dense, overlapping manner, with colors ranging from yellow to red. The most prominent words are 'Marvel' and 'Avengers', both in large, bold, yellow font. Other significant words include 'like', 'cosplay', 'Thanos', 'Infinity', 'got', 'Deadpool', 'Captain', 'Stan', 'America', 'MCU', 'Official', 'War', 'New', 'Thor', 'Man', 'time', 'one poster', 'Daredevil', 'movie', 'made', 'Iron', and 'Deadpool'.

Marvel
Avengers
like
cosplay
Thanos
Infinity
got
Deadpool
Captain
Stan
America
MCU
Official
War
New
Thor
Man
time
one poster
Daredevil
movie
made
Iron

02 - Data Visualization

Word Frequency Horizontal Bar Plot - Harry Potter



1. Harry
2. Potter
3. Hogwarts
4. like
5. one

Logistic Regression Analysis

Cross-Validation Scores: [0.771875 0.80625 0.815625 0.815625 0.80625]				
Mean CV Score: 0.803125				
Training Score: 0.90875				
Testing Score: 0.825				
Accuracy: 0.825				
	precision	recall	f1-score	support
Harry Potter	0.80	0.87	0.83	201
Marvel	0.86	0.78	0.82	199
accuracy			0.82	400
macro avg	0.83	0.82	0.82	400
weighted avg	0.83	0.82	0.82	400

Tips



Logistic Regression and TF-IDF model achieved 82.5% accuracy, demonstrating strong subreddit differentiation with balanced precision and recall, consistent cross-validation, and high training performance.

Accuracy: 0.8

	precision	recall	f1-score	support
Harry Potter	0.79	0.83	0.81	201
Marvel	0.81	0.77	0.79	199
accuracy			0.80	400
macro avg	0.80	0.80	0.80	400
weighted avg	0.80	0.80	0.80	400

Cross-Validation Scores: [0.8175 0.8125 0.84 0.8325 0.845]

Mean CV Score: 0.8295

Training Score: 0.94625

Testing Score: 0.8

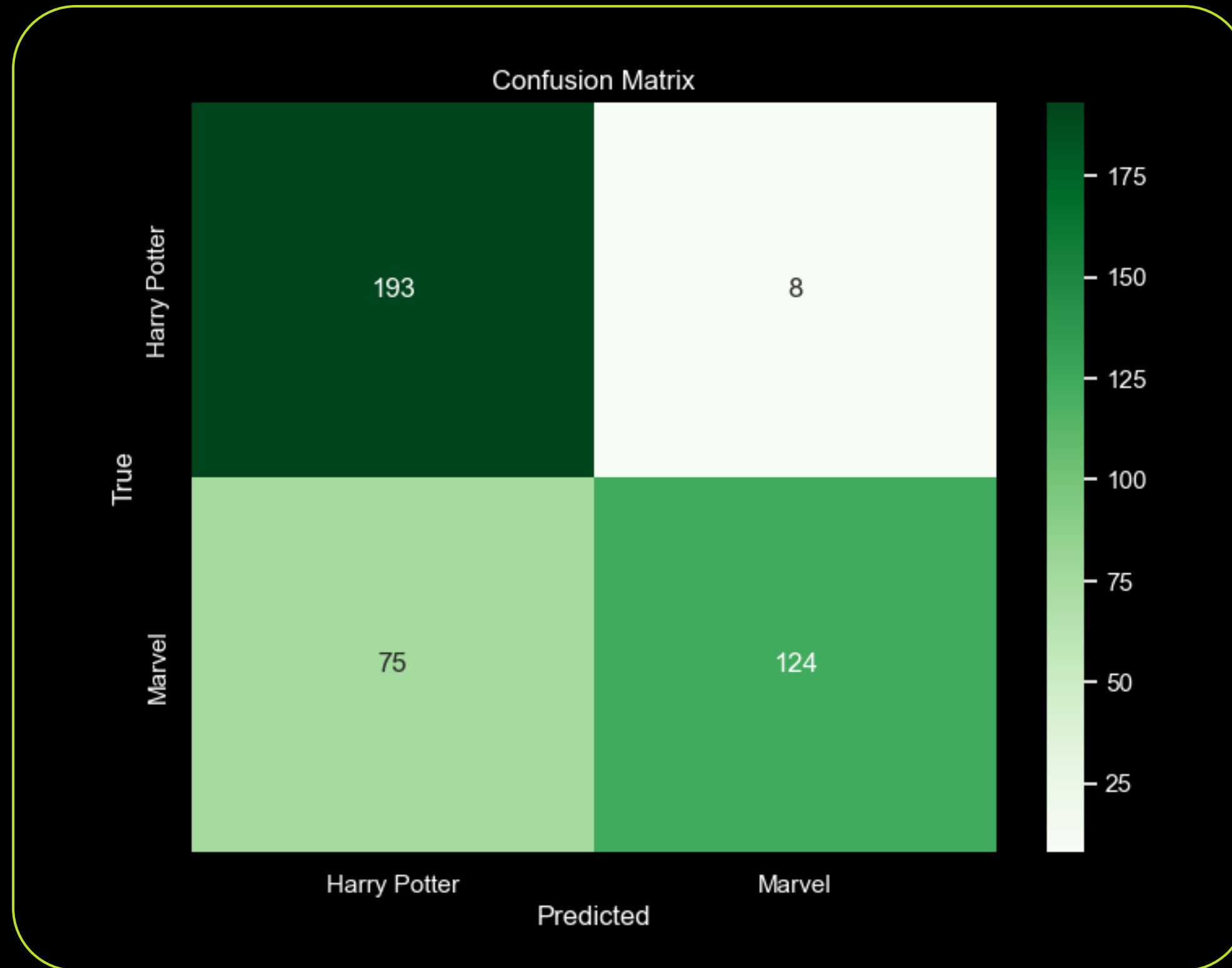
Tips



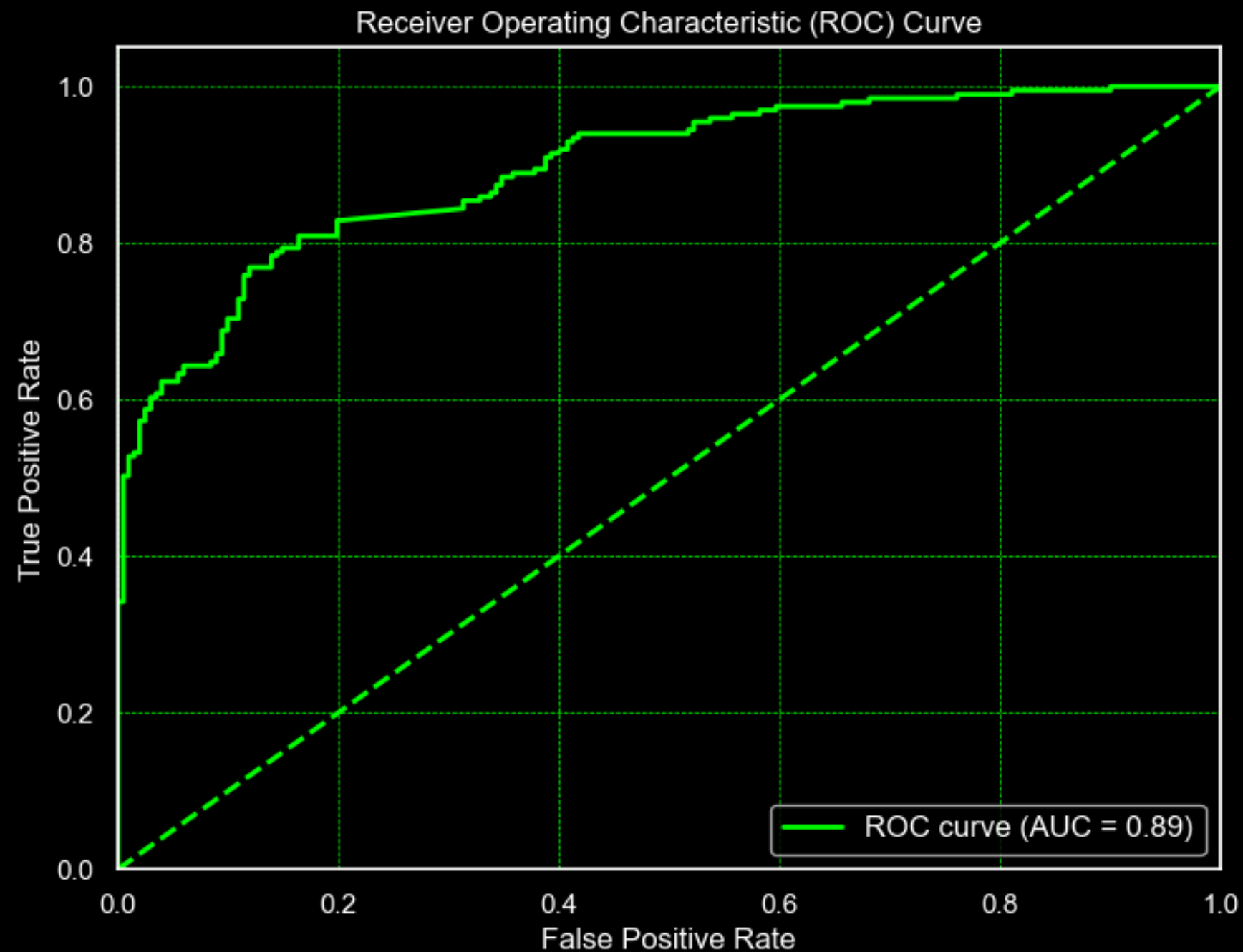
Naive Bayes and TF-IDF model achieved 80% accuracy, demonstrating strong subreddit differentiation with balanced precision and recall, consistent cross-validation, and high training performance.

Naive Bayes Analysis

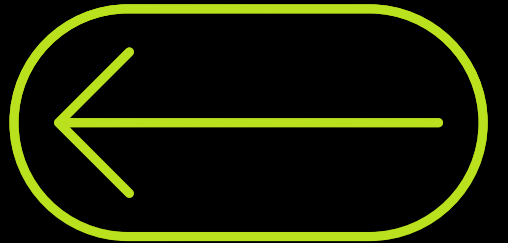
Prediction



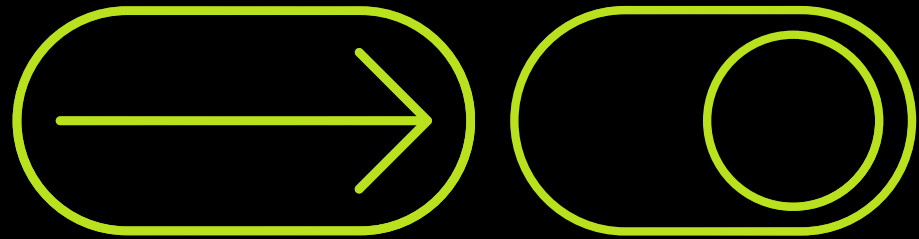
Predictions and Inference



ROC AUC 0.89



04 - Conclusions



1. ***Best Classifier Model: Logistic regression Analysis.***
 - a. ***Title and Content***
2. ***TF-IDF Vectorization (Term Frequency-Inverse Document Frequency)***
3. ***CV = 0.80, Train = 0.91, Test = 0.825***
4. ***ROC AUC = 0.89***

Thanks
